

Unit Inference Rules

Unit inference is conducted on the syntax tree of the numerical expression in each predicate, treating operators as interior nodes and fields/literals as leaf nodes. The inference proceeds in a bottom-up manner, assigning a unit to every node. A unit mismatch is raised whenever an interior node (operator) receives child nodes with incompatible units, making the constraint semantically invalid.

Notation

Symbol	Meaning
$u[n]$	The unit of syntax tree node n
$u[e]^{-1}$	The reciprocal unit of $u[e]$ (e.g., if $u[e] = \text{m}$, then $u[e]^{-1} = \text{m}^{-1}$)
$u[e_1] \cdot u[e_2]$	The product unit (e.g., $\text{m} \cdot \text{s}^{-1}$)
$u[e]^m$	The unit raised to a numeric power m

Rules

1. Leaf Nodes

Leaf nodes are the terminals of the syntax tree: field accesses and numeric literals.

Rule	Formal	Description
U-FIELD	$u[v_k.f] \leftarrow \text{RetrieveFromDef}(f)$	The unit of a field-access leaf node is retrieved from the context definitions
U-LITERAL	$u[c] \leftarrow \text{AskLLM}(c)$	The unit of a numeric literal leaf node is provided by the LLM.

2. Arithmetic Operations

Interior nodes for arithmetic operators infer their unit from their child nodes.

Rule	Formal	Mismatch Cond.	Description
U-NEG	$u[-e] \leftarrow u[e]$	—	A unary negation node inherits the unit of its single child node.
U-ADD	$u[e_1 + e_2] \leftarrow u[e_1]$	$u[e_1] \neq u[e_2]$	Both child nodes must have the same unit; the $+$ node takes that unit.
U-SUB	$u[e_1 - e_2] \leftarrow u[e_1]$	$u[e_1] \neq u[e_2]$	Both child nodes must have the same unit; the $-$ node takes that unit.
U-MUL	$u[e_1 \times e_2] \leftarrow u[e_1] \cdot u[e_2]$	—	The $\times$ node's unit is the product of its two child nodes' units.
U-DIV	$u[e_1 \div e_2] \leftarrow u[e_1] \cdot u[e_2]^{-1}$	—	The $\div$ node's unit is the quotient of its two child nodes' units.
U-MOD	$u[e_1 \% e_2] \leftarrow u[e_1]$	e_2 carries a unit	The $\%$ node takes the unit of its left child node; the right child node must be a pure number (carry no unit).

3. Math Functions

3.1 Unit-Preserving

These function nodes preserve the unit of their child node(s). A unit mismatch is raised when the two child nodes have different units.

Rule	Formal	Mismatch Cond.	Description
U-ABS	$u[\text{abs}(e)] \leftarrow u[e]$	—	An abs node inherits the unit of its child node.
U-MIN	$u[\text{min}(e_1, e_2)] \leftarrow u[e_1]$	$u[e_1] \neq u[e_2]$	Both child nodes must have the same unit; the min node takes that unit.
U-MAX	$u[\text{max}(e_1, e_2)] \leftarrow u[e_1]$	$u[e_1] \neq u[e_2]$	Both child nodes must have the same unit; the max node takes that unit.

3.2 Unit-Restricted Functions

These function nodes impose specific unit requirements on their child nodes. A unit mismatch is raised when the child node's unit is incompatible with the function's mathematical requirement.

Rule	Formal	Mismatch Cond.	Description
U-SIN	$u[\mathbf{sin}(e)] \leftarrow \text{unitless}$	$u[e] \neq \text{rad}$	Child node must have unit rad; function node is unitless.
U-COS	$u[\mathbf{cos}(e)] \leftarrow \text{unitless}$	$u[e] \neq \text{rad}$	
U-TAN	$u[\mathbf{tan}(e)] \leftarrow \text{unitless}$	$u[e] \neq \text{rad}$	
U-ASIN	$u[\mathbf{asin}(e)] \leftarrow \text{rad}$	e carries a unit	Child node must be unitless; function node has unit rad.
U-ACOS	$u[\mathbf{acos}(e)] \leftarrow \text{rad}$	e carries a unit	
U-ATAN	$u[\mathbf{atan}(e)] \leftarrow \text{rad}$	e carries a unit	
U-ATAN2	$u[\mathbf{atan2}(e_1, e_2)] \leftarrow \text{rad}$	$u[e_1] \neq u[e_2]$	Both child nodes must have compatible units; function node has unit rad.
U-EXP	$u[\mathbf{exp}(e)] \leftarrow \text{unitless}$	e carries a unit	Child node must be unitless; function node is unitless.
U-LOG	$u[\mathbf{log}(e)] \leftarrow \text{unitless}$	e carries a unit	
U-LOG10	$u[\mathbf{log10}(e)] \leftarrow \text{unitless}$	e carries a unit	
U-LOG1P	$u[\mathbf{log1p}(e)] \leftarrow \text{unitless}$	e carries a unit	

3.3 Angle Unit Conversion

These function nodes convert between angle units. A unit mismatch is raised when the child node's unit does not match the expected input unit.

Rule	Formal	Mismatch Cond.	Description
U-DEGREES	$u[\mathbf{degrees}(e)] \leftarrow \text{deg}$	$u[e] \neq \text{rad}$	A degrees node converts radians to degrees; child node must have unit rad, and the node itself carries unit deg.
U-RADIANS	$u[\mathbf{radians}(e)] \leftarrow \text{rad}$	$u[e] \neq \text{deg}$	A radians node converts degrees to radians; child node must have unit deg, and the node itself carries unit rad.

3.4 Power and Root Functions

These function nodes compute the unit by raising the child node's unit to a fractional or integer power. No unit mismatch condition applies.

Rule	Formal	Mismatch Cond.	Description
U-POW	$u[\mathbf{pow}(e, m)] \leftarrow u[e]^m$	—	A pow node raises its child node's unit to the numeric power m (e.g., child unit m , $m = 2 \Rightarrow$ node unit m^2).
U-SQRT	$u[\mathbf{sqrt}(e)] \leftarrow u[e]^{1/2}$	—	A sqrt node takes the square-root of its child node's unit (e.g., child unit $m^2 \Rightarrow$ node unit m).
U-CBRT	$u[\mathbf{cbrt}(e)] \leftarrow u[e]^{1/3}$	—	A cbrt node takes the cube-root of its child node's unit.